

Network Topologies

Q1. Diagrams of Star, Bus and Mesh topologies with a real-world example

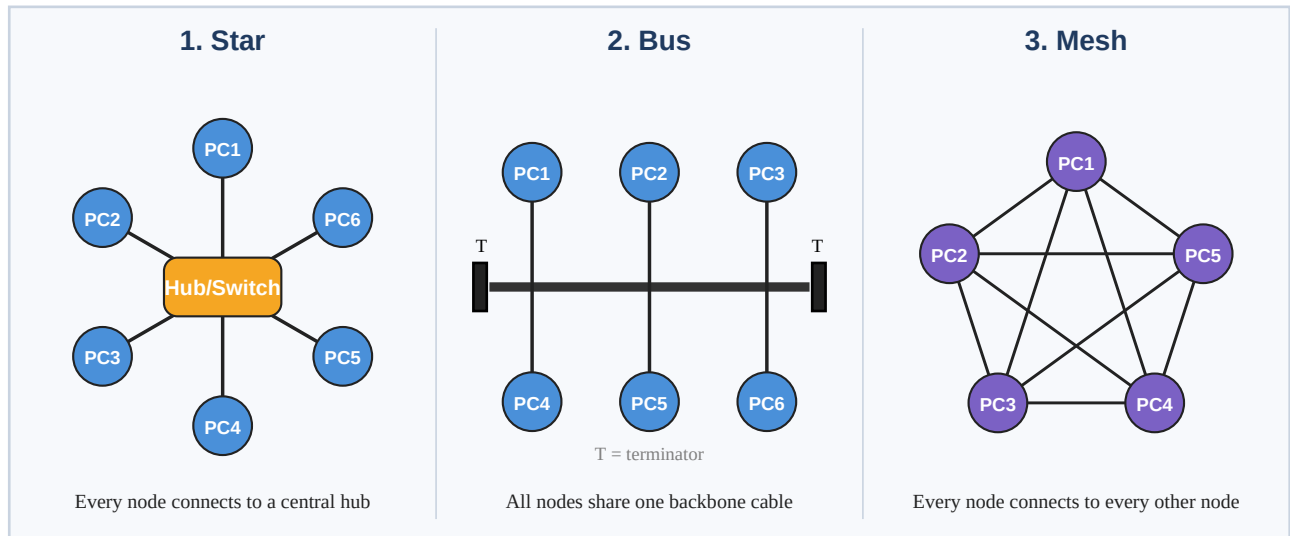


Figure 1: Star, bus and mesh network topologies

Topology	Real-world example	Why
Star	WhatsApp	Every phone connects to WhatsApp's central servers, which receive each message and forward it to the right person. The central point makes control and management easy.
Bus	IPL live streaming (as a broadcast analogy)	One source feed is put on a shared line and every receiver picks it up, just as all nodes listen on one bus cable. Real streaming apps use servers, so this is a concept match.
Mesh	Home mesh WiFi (Google Nest, Jio mesh) and UPI/Paytm bank links	Nodes have several paths to each other, so if one link fails, data takes another route. This suits services that must not go down.

Q2. Comparison of Star, Bus, Ring and Mesh topologies

Topology	Pros	Cons	Best Use Case (with app example)
Star	• Easy to install and manage • Failure of one node does not affect others • Easy to find faults and troubleshoot • Easy to add or remove devices	• Hub/switch is a single point of failure • Needs more cable than bus • Hub cost adds to setup cost • Speed depends on the hub's capacity	Home, college and office WiFi networks, where a router is the centre. Example: using Zomato or Instagram on home WiFi.
Bus	• Cheapest and simplest to set up • Needs very little cable • Easy to extend for small networks • Good for small, temporary networks	• Main cable failure stops the whole network • Performance drops as devices increase • Hard to troubleshoot faults • Data collisions and low security	Very small or temporary networks and vehicle systems (CAN bus). Example: a live IPL broadcast feed shared to many receivers (analogy).
Ring	• Orderly data transfer, no collisions • Every device gets equal access • Performs well under heavy load • Fairly cheap to install	• One broken link or node can break the ring (single ring) • Hard to troubleshoot • Adding a node disturbs the network • Data passes through every node, so it can be slow	ISP and metro fibre backbones. Example: a fibre ring of Jio or Airtel that carries Instagram and YouTube traffic to your city.
Mesh	• Very reliable, with many alternate paths • No single point of failure • Fast direct links, no traffic jams • Easy to isolate faults; good security	• Most expensive: needs many links and ports • Complex to set up and maintain • Many cables or connections needed • Hard to expand as each new node needs many links	Critical networks that must never go down. Example: Paytm/UPI payment links between banks, and mesh WiFi at home.

Q3. Network topology for a multiplayer mobile game (like PUBG / Free Fire)

I would choose a **star topology (client-server model)**, with game servers placed in several regions and the servers themselves linked in a mesh for backup. Every player's phone connects to one central game server, which keeps the match state in sync and helps prevent cheating. **Speed:** a nearby regional server keeps ping low and gameplay smooth. **Reliability:** one failed phone does not affect the other players, and backup servers can take over if a server fails. **Cost:** each player needs only one connection and no device-to-device links, so it is far cheaper than a full mesh (pure peer-to-peer) design, though servers must be paid for and maintained.

Q4. Network for a small warehouse with 8 computers

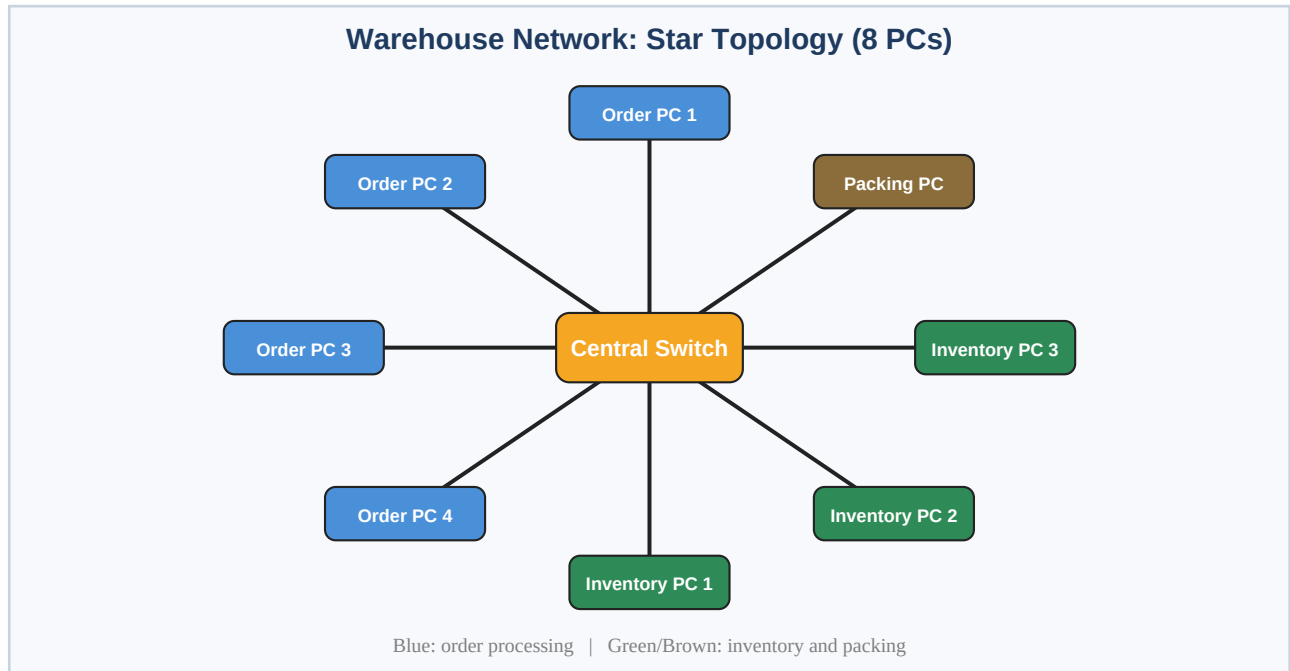


Figure 2: Recommended star topology for the warehouse

Recommendation: Star topology with one central switch. It is low cost because 8 PCs need only one switch and simple cables, unlike the many links of a mesh. It is easy to troubleshoot, since a fault in one PC's cable only affects that PC while order processing and inventory updates continue on the others. It is also easy to expand: more PCs can be added by plugging into the switch or by adding a second switch later.